OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

Under deceleration conditions, the PCM adjusts the Idle Air Control (IAC) motor to maintain a desired MAP value. Under idle conditions, the PCM adjusts the IAC motor to maintain a desired engine speed.

REMOVAL

The Vehicle Speed Sensor (VSS) is located on the speedometer pinion gear adapter. If equipped with 4WD, this adapter is located on the transfer case extension (left side) (Fig. 100). If equipped with 2WD, this adapter is located on the extension housing of the transmission (left side).

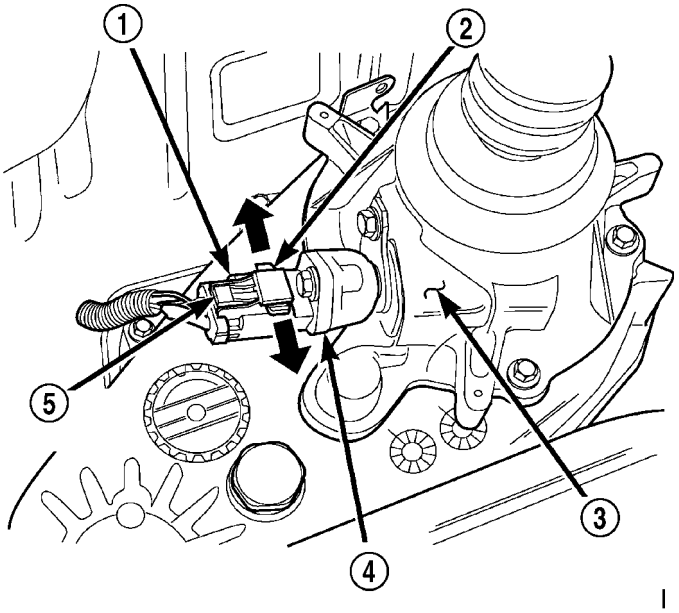

Fig. 100 VSS Location

1 - SENSOR ELECTRICAL CONNECTOR
2 - SLIDE TAB
3 - 4WD TRANSFER CASE EXTENSION
4 - VEHICLE SPEED SENSOR
5 - RELEASE LOCK

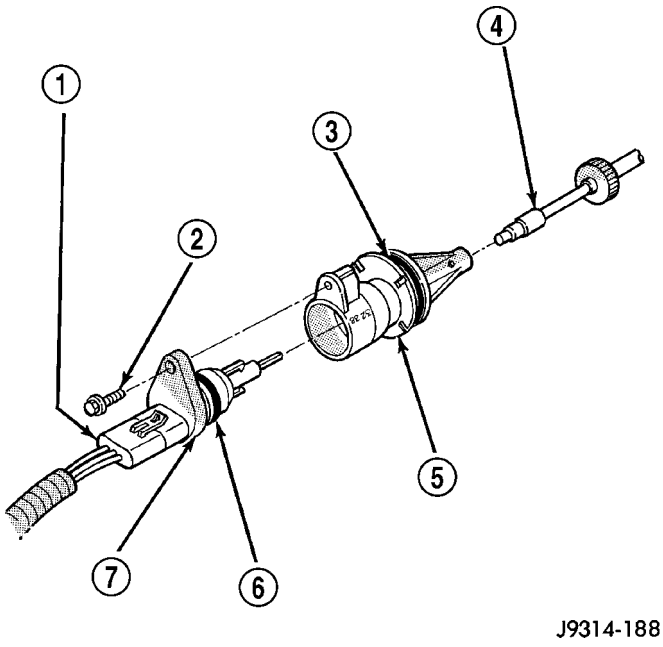

Fig. 101 VSS Removal/Installation

1 - ELECTRICAL CONNECTOR
2 - SENSOR MOUNTING BOLT
3 - O-RING
4 - SPEEDOMETER PINION GEAR
5 - SPEEDOMETER PINION GEAR ADAPTER
6 - O-RING
7 - VEHICLE SPEED SENSOR

(1) Raise and support vehicle.

(2) Disconnect electrical connector from sensor by pushing slide tab (Fig. 100). After slide tab has been positioned, push in on secondary release lock (Fig. 100) on side of connector and pull connector from sensor.

(3) Remove sensor mounting bolt (Fig. 101).

(4) Remove sensor (pull straight out) from speedometer pinion gear adapter (Fig. 101). Do not remove gear adapter from transmission.

INSTALLATION

(1) Clean inside of speedometer pinion gear adapter before installing speed sensor.

(2) Install sensor into speedometer gear adapter and install mounting bolt. Before tightening bolt, verify speed sensor is fully seated (mounted flush) to speedometer pinion gear adapter.

(3) Tighten sensor mounting bolt to 2.2 N·m (20 in. lbs.) torque.

(4) Connect electrical connector to sensor.